

Using Computer Algebra in Mathematics for Engineers

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Abstract

Computer algebra systems (CAS) have improved the mathematical work of engineers. The systems are used for numerical computations as well as for algebraic manipulations of equations. Moreover, the powerful graphical capabilities and the easy use of the graphics are applied to display complicated functions and technical results. The techniques in hand calculations are trusted into the background in favor of the systematic approach in mathematics and of the exciting modelling of realistic systems. This exciting aspect has been taken up and the CAS Maple was included in the education of engineers. Mathematical concepts are motivated in a clear and vivid manner by the use of the visualization and animation capabilities of Maple.

In this paper the principal concept and the application of Maple in engineering education will be demonstrated in various examples:

- Lengthy and abstract topics like the convergence of Fourier series to a given function are discussed.
- The visualization of the wave equation in case of a vibrating string is performed.
- Eigenvectors can be identified geometrically by showing an animation of a rotating vector.
- Finally, the oscillations of an idealized skyscraper are computed to visualize the meaning of eigenvalues and eigenvectors, physically.

For each of these examples a worksheet can be used interactively.

Keywords

Mathematics for Engineers, Maple, Mathematical Concepts, Visualization